

Business Rule Testing (<i>not testing CRUD here, see next section</i>)

EP TCIs

- **name**
- **gpa**

TCI	Parameter	Range of Inputs
EP1	gpa	0..4.0
EP2	name	1..255

EP Test Cases (I am not covering values outside of the 2 EPs, as they will be caught in BVA testing)

Test Case ID	TCI Covered	gpa	name	Exp.
TC1.1	EP1[2]	2.5	Jim Hopper	Accepted as valid

BV TCIs

- **name**
- **gpa**

TCI	Parameter	Boundary Value
BV1*	gpa	-1.0
BV2		0
BV3		1.0
BV4		2.0
BV5		3.99

BV6		4.0
BV7*		4.1
BV8*	name	NULL
BV9		1
BV10		2
BV11		254
BV12		255
BV13*		256

BV Test Cases (grouping test cases together that yield the same results to optimize, [] = same outcome)

Test Case ID	TCI Covered	gpa	name	Exp.
BVA1.1*	BV1	-1.0	Jim Hopper	Exception: gpa outside of range
BVA1.2	BV2[3][4][5][6]	0.0	Jim Hopper	Accepted as valid
BVA1.3*	BV8[13]	3.5	Some name over 255 ch. OR no name :)	Exception: name outside of range
BVA1.4	BV9[10][11][12]	4.0	Jim Hopper	Accepted as valid
BVA1.5*	BV6	4.1	Jim Hopper	Exception: gpa outside range

DT Testing (not retesting cases from EP/BVA)

Cause

- Course Full?

Effect

- False = Can add student (rule 1)
- True = Cannot add student (rule 2)

Test Case ID	TCI Covered	gpa	name	Course Full?	Exp
DT1.1	DT1	3.5	Jim Hopper	F	Can add student
DT1.2	DT2	3.5	Jim Hopper	T	Cannot add student

CRUD Process Testing (*illustrated as state transitions*)

Student States

- Student does not exist yet
- Create student
- Update student info
- Delete a student

Student Transitions (like a TCI)

API Command	Current State	Resulting State
POST (create student)	Does not exist	Student created
PUT (update student)	Student created	Student info updated
DELETE (delete student)	Student created/updated	Student deleted
GET (get student)	Student created/updated	Student info read

Student Transitions (Test Cases)

Test Case ID	Current State	API Command	Resulting State	Exp.
ST1.1	Student does not exist	POST	Created	Student created
ST1.2	Student created	GET	Created	Student info returned
ST1.3	Student created	PUT	Updated	Student info updated
ST1.4	Student created	DELETE <code>/students/{id}</code>	Deleted	Student removed
ST1.5*	Student does not exist	PUT	Student does not exist	Error
ST1.6*	Student removed	GET	Student removed	Error

Course Transitions (like a TCI)

- Course does not exist yet
- Create course
- Update course info
- Delete a course

API Command	Current State	Resulting State
POST (create course)	Does not exist	Course created
PUT (update course)	Course created	Course info updated
DELETE (delete course)	Course created/updated	Course deleted

GET (get all courses)	Course created/updated	Course info returned
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Course Transitions (Test Cases)

Test Case ID	Current State	API Command	Resulting State	Exp.
ST2.1	Does not exist	POST	Created	Course created
ST2.2	Course(s) created	PUT <code>/courses/{id}</code>	Updated	Course info updated
ST2.3	Course(s) created	DELETE <code>/courses/{id}</code>	Removed	Course deleted
ST2.4	Course(s) created	GET <code>/courses/{id}</code>	Retrieved	Course(s) retrieved

Student + Course State Transitions (like a TCI)

- Add student to an existing course with an empty roster
- Add student to an existing course with students
- Delete student from an existing course with only that student
- Delete student from an existing course with many students

API Command	Current State	Resulting State
POST (add student to a course)	Empty Course	Has Student(s)
POST (add student to a course)	Has student(s)	Full

DELETE (remove student from course)	Has student	Empty
DELETE (remove student from course)	Full	Has student(s)

Student + Course State Transitions (Test Cases)

Test Case ID	Current State	API Command	Resulting State	Exp.
ST2.5	Empty course	POST /{courseId}/students/{studentId}	Has student(s)	Valid operation
ST2.6	Has student(s)	POST /{courseId}/students/{studentId}	Full	Valid operation
ST2.7*	Full course	POST /{courseId}/students/{studentId}	Full course	Invalid operation
ST2.8	Has student(s)	DELETE /{courseId}/students/{studentId}	Empty/Has student(s)	Valid operation
ST2.9*	Empty course	DELETE /{courseId}/students/{studentId}	Empty	Invalid operation